32-bit microcontrollers

HC32L07x_F072 Series Notes

Application Notes

Rev1.04 February 2025

Applicable objects

Product range	Product Model	Product range	Product Model
L Series	HC32L072 HC32L073	F Series	HC32F072

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1 Summary

This document mainly introduces the precautions and workarounds for using the HC32L07x/ HC32F072 series chips.

2 Notes on HC32L07x/ HC32F072 Series

Table 2-1 Statistics of precautions

Precautions	Specific precautions
2.1 System and clock considerations	2.1.1 VCAP pin
	2.1.2 DVCC and AVCC decoupling capacitors
	2.1.3 BOOT pin
	2.1.4 NRESET pin
	2.1.5 SWD interface failure reasons
	2.1.6 UART pins in ISP mode
	2.1.7 I/O port voltage
	2.1.8 Chip encryption
	2.1.9 External crystal oscillator circuit
	2.1.10 XTH crystal selection
	2.1.11 XTH configuration
	2.1.12 XTL configuration
	2.1.13 RCH frequency switching
	2.1.14 External clock input
2.2 Reset controller	2.2.1 Reset flags after power-on
	2.2.2 Peripheral modules reset
2.3 Flash	2.3.1 Flash erase and write functions address range
	2.3.2 Flash instruction fetch waiting cycle
	2.3.3 Flash operation time parameters
2.4 Ultra-low power consumption	2.4.1 ADC handling
	2.4.2 I/O handling
	2.4.3 SWD handling
	2.4.4 XTH handling
	2.4.5 PLL handling
2.5 ADC	2.5.1 Analog power pin
	2.5.2 Operating voltage and sampling rate
	2.5.3 Sampling rate and input impedance relations
	2.5.4 BGR needs to be enabled before using ADC
	2.5.5 ADC signal voltage
	2.5.6 Improving ADC sampling accuracy
	2.5.7 ADC 1Msps sampling rate precautions
2.6 LCD	2.6.1 High driving capability displays
	2.6.2 COM2/COM3 remapping
	2.6.3 BOOT pin restrictions

Precautions	Specific precautions
2.7 BGR	2.7.1 BGR related modules
2.8 I2C	2.8.1 I2C 1M rate
2.9 SPI	2.9.1 Bit0 and bit1 of SPI_CR2 register
2.10 LVD	2.10.1 LVD input connection
2.11 WDT	2.11.1 Stop problem after WDT is turned on
	2.11.2 10K WDT clock deviations
2.12 GPIO	2.12.1 SWD pins multiplexing
	2.12.2 GPIO protection
	2.12.3 GPI input filtering circuits
	2.12.4 Internal pull-up and pull-down resistors
	2.12.5 BOOT pin GPIO number
2.13 DMA	2.13.1 Setting GPIO using DMA
2.14 VC	2.14.1 Resistor voltage divider function
	2.14.2 Internal filter time
2.15 OPA	2.15.1 Zero calibration function missing
2.16 USB	2.16.1 USB_DP and USB_DM pins
2.17 Timer	2.17.1 M23CR register
2.18 RTC	2.18.1 Reading and writing the RTC counter register

2.1 System and clock considerations

2.1.1 VCAP pin

■ Application Notes

a. VCAP is the core voltage filter pin and must be grounded through a capacitor. It cannot be used for other functions or left floating.

Note: For the minimum allowable capacitance of each series, please refer to the data sheet of each series.

b. Place the capacitor as close to the VCAP pin as possible.

2.1.2 DVCC and AVCC decoupling capacitors

■ Application Note

Each DVCC and AVCC pin needs to be equipped with a decoupling capacitor, as close as possible to the corresponding power pin. The specific capacitance value can be determined according to the actual situation of the board.

2.1.3 BOOT pin

■ Application Notes

a. When used as user mode, the BOOT pin must be a low level input when powered on or reset. The MCU detects the BOOT pin status when powered on or reset. When the input is low level, it directly enters the user main program, and when the input is high level, it enters the boot program (used for ISP mode (UART) burning).

Note: If the BOOT pin is not low level when powered on or reset, the user main program cannot be entered.

b. It is not recommended to use the BOOT pin for other functions.

c. If the BOOT pin is to be used as a GPIO, note that when powered on or reset, the BOOT peripheral circuit can keep the pin in a pull-down state.

2.1.4 NRESET pin

■ Application Note

Inside the MCU, NRESET has a weak pull-up, but from the perspective of system reliability, it is still recommended to refer to the data sheet for the recommended circuit external reset circuit.

2.1.5 SWD interface failure reasons

■ Application Notes

a. The user main program enters deep sleep mode as soon as it is powered on.

b. After burning the user program and encrypting it, the SWD cannot be connected after the burning is completed and it is powered on or reset again.

c. The chip cannot be connected and downloaded using SWD again due to an incorrect program.

d. Immediately after reset, the SWD interface is changed to the GPIO interface, causing the emulator or programmer to fail to connect.

■ Recovery methods

a. Use the online programming tool to erase the user program on the MCU using the chip erase method.

b. Use the offline programming tool to configure a normal example, select chip erase and no encryption, and then download the program once.

c. For the above three situations a, c, and d, during the debugging stage: you can add a delay of more than 1s at the beginning of main so that you can burn the program again through the SWD interface after the next power-on or reset; or at the beginning of main, detect the external level of a certain IO input to determine whether to continue to execute the subsequent program.

2.1.6 UART pins in ISP mode

■ Application Notes

a. The ISP pin and SWD pin of this series are multiplexed.

For details, please refer to the user manual of the online programmer or offline programmer.

2.1.7 I/O port voltage

■ Application Notes

a. All I/O voltages cannot be greater than VCC.

Note: Engineers like to use PC+USB-UART+serial debugging assistant to view the data sent by UART during debugging. At this time, if the UART level is greater than the MCU power supply voltage VCC, there will be problems.

b. When I/O is powered on before Vcc, or when the I/O pin is powered but Vcc is not powered, the voltage injected from this I/O pin may also cause the MCU to run, but it may not run normally. If this situation cannot be avoided, it is recommended to connect a suitable resistor in series with this I/O to limit the voltage and current injected into the MCU to prevent the MCU from running under such circumstances.

2.1.8 Chip encryption

■ Application Notes

a. After the chip is encrypted, it cannot be simulated or downloaded. It is necessary to perform chip erase operation through the ISP interface using an online or offline programmer before resuming simulation and downloading.

b. The encryption and decryption of this series is limited in the number of times. After 64 cycles of encryption and decryption, the chip cannot be encrypted or decrypted again.

2.1.9 External crystal oscillator circuit

■ Application Notes

a. The external crystal oscillator unit and matching capacitors should be as close to the chip as possible.

b. The external crystal oscillator signal line should be as short as possible. The line width should not be too thin, and the thinnest should not be less than the width of the chip pin.

c. There should be a complete ground plane in the layer adjacent to the crystal oscillator local circuit.

d. A ground wire should be used as a guard ring around the external crystal oscillator. The ground ring wire needs to be fully grounded (more than ground holes) to reduce the mutual interference between the external crystal oscillator signal and other signals.

e. The crystal oscillator circuit should pay attention to the cleanness of local signals and avoid external interference. Try not to run wires near the crystal oscillator circuit or in adjacent layers, especially high-speed wires, power wires, clock wires, etc.

f. In harsh application environments, such as humid environments, in order to reduce the oscillation problem caused by leakage, it is necessary to add a coating, such as three-proof paint, to the PCB board in the external crystal oscillator area.

2.1.10 XTH crystal selection

If you choose a passive crystal oscillator, it is recommended to choose a low ESR 16MHz~32MHz crystal.

2.1.11 XTH configuration

■ Application Notes

a. The XTHI/XTHO pins should be set to analog functions in the program.

b. After the XTH stable flag is detected in the program, it is necessary to wait at least 10ms before using the XTH clock as the system clock or PLL clock source.

c. The external crystal oscillator should be matched with a suitable matching capacitor and have a suitable driving capability. If necessary, a current limiting resistor can be connected in series to the XTH_OUT pin.

2.1.12 XTL configuration

■ Application Notes

a. In the program, the XTLI/XTLO pins should be set to analog functions.

b. If power consumption permits, XTL should use high drive capability, but the higher the drive capability, the greater the power consumption.

2.1.13 RCH frequency switching

■ Application Notes

Solution 1: When switching RCH, you must follow the process of adjusting the HCLK frequency first, then gradually increasing it, or gradually decreasing it, and then adjusting the HCLK frequency. It cannot be done in one step.

Solution 2: First switch to RCL as the system clock, then change the RCH frequency, and then switch back to RCH as the system clock.

Otherwise, it may affect the operation of the user program.

For details, please refer to the system controller chapter and related application notes of the reference manual of each series.

External clock input

■ Application Notes

a. When using XTH external input (from XTHI), the XTHI pin needs to be set as digital input and SYSCTRL0.XTHEN is 1. At the same time, the XTHO pin is prohibited from being used as GPIO and needs to be configured as analog state.

When using XTL external input (from XTLI), the XTLI pin needs to be set as digital input and SYSCTRL0.XTLEN is 1. At the same time, the XTLO pin is prohibited from being used as GPIO and needs to be configured as analog state.

2.2 Reset controller

2.2.1 Reset flags after power-on

■ Application Note

The bits of the reset flag register other than POR5V and POR15V are in an undefined state after power-on and need to be initialized and cleared by software.

2.2.2 Peripheral modules reset

■ Application Note

To use the peripheral module reset control register to reset a peripheral module, you need to write the corresponding bit control bit to 0 first and then to 1.

2.3 Flash

2.3.1 Flash erase and write functions address range

■ Application Note

Flash erase and write functions need to be within the Flash address range of 0~32K and cannot exceed it, or run in RAM, otherwise errors will occur when performing erase and write operations.

For details, refer to the application note: FLASH operation instructions.

2.3.2 Flash instruction fetch waiting cycle

■ Application Note

When HCLK ≤ 24MHz: FLASH_CR.WAIT=0

When 24MHz ＜ HCLK ≤ 48MHz: FLASH_CR.WAIT=1

2.3.3 Flash operation time parameters

■ Application Note

The configuration of the FLASH erase and write related time parameter register is strongly related to HCLK and should comply with the reference manual requirements.

2.4 Ultra-low power consumption

2.4.1 ADC handling

■ Application Note

Before entering ultra-low power consumption, you need to turn off BGR first, then turn off ADC_EN, and then turn off the ADC peripheral clock to further reduce the power consumption in the ultra-low power consumption state.

2.4.2 I/O handling

■ Application Notes

a. For chips that are not packaged with the maximum PIN, refer to the maximum package and set the unpackaged pins to a fixed digital level (such as pull-up or pull-down input) to ensure that there is no leakage. For example: the 48pin of L136 should also be set to 64pin, and the unpackaged pins should be set as above.

b. The unused floating pins should also be processed as above.

c. Analog signal pins such as ADC that have been connected to external circuits do not require additional processing when in standby mode.

d. For output pins, for external circuits, avoid generating current pull or current sink during standby mode.

e. If you want to reduce standby power consumption, you can also refer to the above processing.

f. If the USB module is not used for L07x/ F07x, the USB-related pins can be grounded to obtain lower power consumption.

2.4.3 SWD handling

■ Application Note

After entering ultra-low power mode, the SWD interface will be invalid and needs to be re-awakened or reset before entering ultra-low power mode to burn the code or erase the original code.

Therefore, it is recommended that during the debugging stage, before entering DeepSleep, you can use a certain length of delay or while query to wait for the valid level to delay the time to enter DeepSleep again, so that you can burn again. Or use the serial port mode of the online or offline programmer to replace or erase the original program.

2.4.4 XTH handling

■ Application Note

When the external high-speed (passive) clock XTH is used as the system clock, before entering DeepSleep mode, you need to set the internal high-speed clock function after wake-up (such as the WAKEUPBYRCH or WAKEUPCLK control bit of the SYSCTRL_CRx register). After wake-up, the system runs under the internal high-speed clock, and then enable and switch XTH according to the steps required by the reference manual.

2.4.5 PLL handling

■ Application Note

When PLL is used as the system clock, before entering DeepSleep mode, you need to switch the system clock to the internal high-speed clock (such as RCH) and turn off BGR. After waking up, the system runs at the internal high-speed clock, and then you need to use software to switch the system clock to the PLL clock.

2.5 ADC

2.5.1 Analog power pin

■ Application Notes

a. AVCC/DVCC/AVSS/DVSS are recommended to be separated to improve ADC accuracy.

b. AVCC needs to be the same as DVCC voltage.

c. When using the internal reference voltage, AVCC must be greater than the internal reference voltage +0.3V.

d. The external reference voltage REF must not be greater than AVCC.

2.5.2 Operating voltage and sampling rate

■ Application Note

The ADC sampling rate has requirements on the operating voltage. For details, please refer to the ADC section of the reference manual.

2.5.5 ADC signal voltage

■ Application Note

The input voltage of the ADC pin cannot exceed the ADC reference voltage.

2.5.6 Improving ADC sampling accuracy

■ Application Notes

a. If a more accurate reference voltage can be provided externally, it is recommended to use an external reference voltage.

b. Reduce the sampling signal impedance.

c. For signals with small changes, try to reduce the sampling rate if the sampling time is met.

d. When the signal impedance is appropriate or the signal strength is sufficient, avoid using the internal follower BUF.

e. Try to turn off unused digital circuits at the ADC sampling moment.

f. Try not to reuse the same group of I/O or I/O with ADC function as high-frequency output or input signal pins.

2.5.7 ADC 1Msps sampling rate precautions

■ Application Note

When the ADC uses a 1 Msps sampling rate, it is necessary to comprehensively consider factors such as the driving capability of the sampled signal and the external input impedance. You can refer to the relevant description in the data sheet.

When the driving capability of the sampled signal is weak, the external input impedance is large, and the sampling rate is too high, it will cause voltage spikes on some channels when scanning channels with large voltage differences before and after sampling, or inaccurate ADC sampling values during single or scan sampling. At this time, it is necessary to consider reducing the sampling rate, enhancing the driving capability of the sampled signal, or reducing the input impedance.

2.6 LCD

2.6.1 High driving capability displays

■ Application Notes

a. For LCD applications with high driving capability requirements, or applications with poor display effects using 8COM, it is recommended to use the external resistor mode, which can provide stronger driving capability.

b. For LCD applications with high power consumption requirements, the external capacitor mode can be considered.

2.6.2 COM2/COM3 remapping

■ Application Note

The COM2/COM3 of the HC32L073PATA-LQFP100 package is different from the pins of other series MCUs, and the configuration is slightly different. It needs to be remapped to PF6 and PA15.

2.6.3 BOOT pin restrictions

■ Application Note

The I/O of the BOOT pin is a SEG30 function. If other SEG pins are sufficient, it is recommended not to use the BOOT pin as a SEG function.

2.7 BGR

2.7.1 BGR related modules

■ Application Notes

a. When using ADC/OPA/PLL, VC high power consumption mode, you need to turn on the BGR module first. After turning on the BGR, it takes a period of stabilization time. Please refer to the reference or data manual for the specific time.

2.8 I2C

2.8.1 I2C 1M rate

■ Application Note

At high speed (1M rate), the time of THD.STA and TSU.STO does not conform to the standard protocol, and the time redundancy at other rates is also small. Please refer to the data sheet of each model for details.

2.9 SPI

2.9.1 Bit0 and bit1 of SPI_CR2 register

■ Application Note

Please refer to the reference manual. Bit0 and bit1 of SPI_CR2 cannot be written to 0 and must remain 1.

2.10 LVD

2.10.1 LVD input connection

■ Application Note

The LVD input is connected to the negative phase of the LVD comparator, and the reference voltage is connected to the positive phase of the LVD comparator. Therefore, the rising edge trigger described in the reference manual is when the monitored voltage changes from higher than the threshold voltage to lower than the threshold voltage.

2.11 WDT

2.11.1 Stop problem after WDT is turned on

■ Application Note

The WDT of the MCU models listed in the list cannot be stopped after it is started, so after entering the ultra-low power state, it is necessary to wake up and feed the dog regularly.

2.11.2 10K WDT clock deviations

■ Application Note

The WDT of the MCU model listed in the table uses a dedicated 10K clock. When using it, it is recommended to clear the WDT before the count value of the WDT counter reaches half of the overflow value.

2.12 GPIO

2.12.1 SWD pins multiplexing

■ Application Notes

a. The SWD pin is in SWD function by default after power-on or reset. If you want to change it to GPIO function, you need to set the switch in the system controller SYSCTRL.CR1. Note that you must operate the write protection register before switching. The driver library of each model provides interface functions for operation.

b. Once changed to GPIO, the SWD function cannot be used. Therefore, in the code development stage, it is best to add software delay before switching. Do not change it immediately after power-on or reset, otherwise the next SWD mode burning or simulation will encounter the problem of being unable to download code.

c. Changing SWD to I/O does not encrypt the chip and cannot prevent others from reading the code on the chip. To achieve true chip encryption, you need to select the [Chip Encryption] function when configuring the offline programmer or using the online programming tool to burn.

2.12.2 GPIO protection

■ Application Note

If the MCU's I/O is used as an external interface pin, and the pin is directly connected to the interface, there is a possibility of live plugging and unplugging, such as UART, it is necessary to add a protection circuit between the interface and the MCU to avoid damaging the MCU.

2.12.3 GPI input filtering circuits

■ Application Notes

a. When the pins of the series MCU are used as GPIO digital input functions, they do not contain input hardware filtering functions. When using them, users need to consider whether to configure hardware input filtering circuits outside the MCU according to the situation.

b. When I/O is multiplexed as other IP function pins, please refer to the reference manual of each series model to see whether there is filtering.

2.12.4 Internal pull-up and pull-down resistors

The internal pull-up and pull-down resistors are MOS resistors. Under 3.3V and 5V power supply conditions, there are differences in resistance. If users need precise pull-up and pull-down, or pull-up and pull-down with smaller resistance, they need to consider adding pull-up or pull-down resistors outside the chip.

2.12.5 BOOT pin GPIO number

■ Application Notes

a. The GPIO pin number corresponding to the BOOT pin of this series of MCUs is PF11.

b. It is not recommended to use the BOOT pin for other functions. If used for other functions, pay attention to the peripheral circuit of BOOT to keep the pin in the pull-down state when powering on or resetting.

c. When processing the floating IO state before entering ultra-low power consumption, pay attention to the processing method of the BOOT pin. The BOOT pin has been connected to the ground by an external resistor.

2.13 DMA

2.13.1 Setting GPIO using DMA

■ Application Note

GPIO input/output value register has two bus control modes: FAST IO and AHB mode. If you want to output data from GPIO via DMA, you need to set the GPIO bus to AHB bus (GPIO_CTRL2.ahb_sel).

2.14 VC

2.14.1 Resistor voltage divider function

■ Application Note

This series of MCUs has more than 2 VC comparators, but only VC0 and VC1 support resistor voltage division as the negative input function.

2.14.2 Internal filter time

■ Application Note

The internal filtering function of the VC of this series of MCU uses the counting clock RC150 K clock, which has low precision. Therefore, the filtering time set by the register may have a large deviation, which needs to be considered when designing. If necessary, the input signal can be matched with an external filtering circuit.

2.15 OPA

2.15.1 Zero calibration function missing

■ Application Note

This series of MCUs with OPA function does not have a zero calibration function.

2.16 USB

2.16.1 USB_DP and USB_DM pins

■ Application Note

This series of MCUs has USB function, and its USB pins are dedicated pins and cannot be reused as GPIO.

2.17 Timer

2.17.1 M23CR register

■ Application Notes

The DIR bit in the M23CR register of the general timer TIM0/1/2/3 can only be written in sawtooth mode, and is read-only and write-prohibited in other modes.

Note: The ARM processor uses the "Load-Store architecture", so the write operation to any bit of the register will eventually be manifested as a write operation to all bits of the register when the CPU executes the write instruction.

When using it, one of the following principles must be followed:

1) The M23CR register is only configured during initialization, and do not write at other times.

2) In non-sawtooth mode, after the timer is enabled, if any write operation is performed on other bits of this register except DIR, the count overflow point and underflow point must be avoided.

2.18 RTC

2.18.1 Reading and writing the RTC counter register

Only applicable to L07x series:

■ Application Note

When using the method of setting RTC_CR1.WAIT to read and write the RTC count register, you must avoid the occurrence of alarm interrupts and periodic interrupts. It is recommended to operate in the RTC interrupt function and avoid being interrupted by other high-priority interrupts.

Note: ARM processors use the "Load-Store architecture", so the write operation of any bit of the register will eventually be manifested as a write operation of all bits of the register when the CPU executes the write instruction.

Revision History

Version	Revision Date	Revision content
Rev1.00	2023/04/28	Initial release.
Rev1.01	2023/05/25	1) Added "Timer" chapter. 2) Changed "THO.STA" to "THD.STA" in "I2C" chapter.
Rev1.02	2024/04/19	1) Added "System and Clock Notes" chapter: Added notes on using external input clock. 2) Added "ADC" chapter: Added notes on using ADC 1 Msps sampling rate. 3) Modified "BGR" chapter: Modified the description of BGR related module notes.) 4) "GPIO" chapter: Deleted the note "When the external crystal is not in use, the pin can be used as GPIO". 5) "VC" chapter: Added the note about the use of the filter time inside VC. 6) Added the note about RTC_CR1.WAIT in the "RTC" chapter.
Rev1.03	2024/09/02	1) System and Clock section: Modify the recommended frequency range of XTH when using passive crystal oscillator. 2) Ultra-low power section: Add XTH as system clock, which requires entering DeepSleep mode; add PLL as system clock, which requires entering DeepSleep mode.
Rev1.04	2025/02/20	In the "System and Clock Considerations" section, modified the description of "Precautions for Using External Input Clocks".